

MENG-465 – Design Project 1

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Cold Plate Heat Exchanger – CFD Project 1

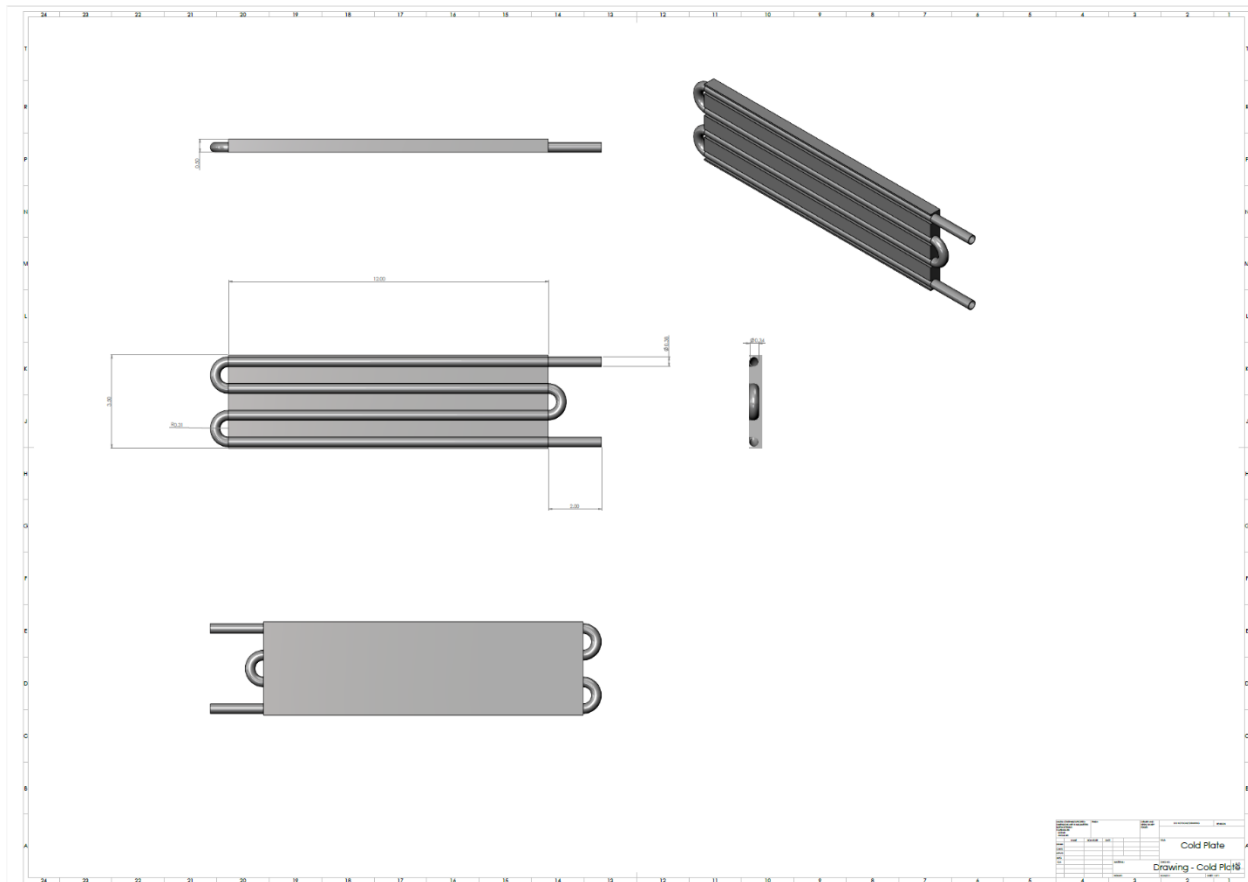


Figure 1. Cold Plate Drawing

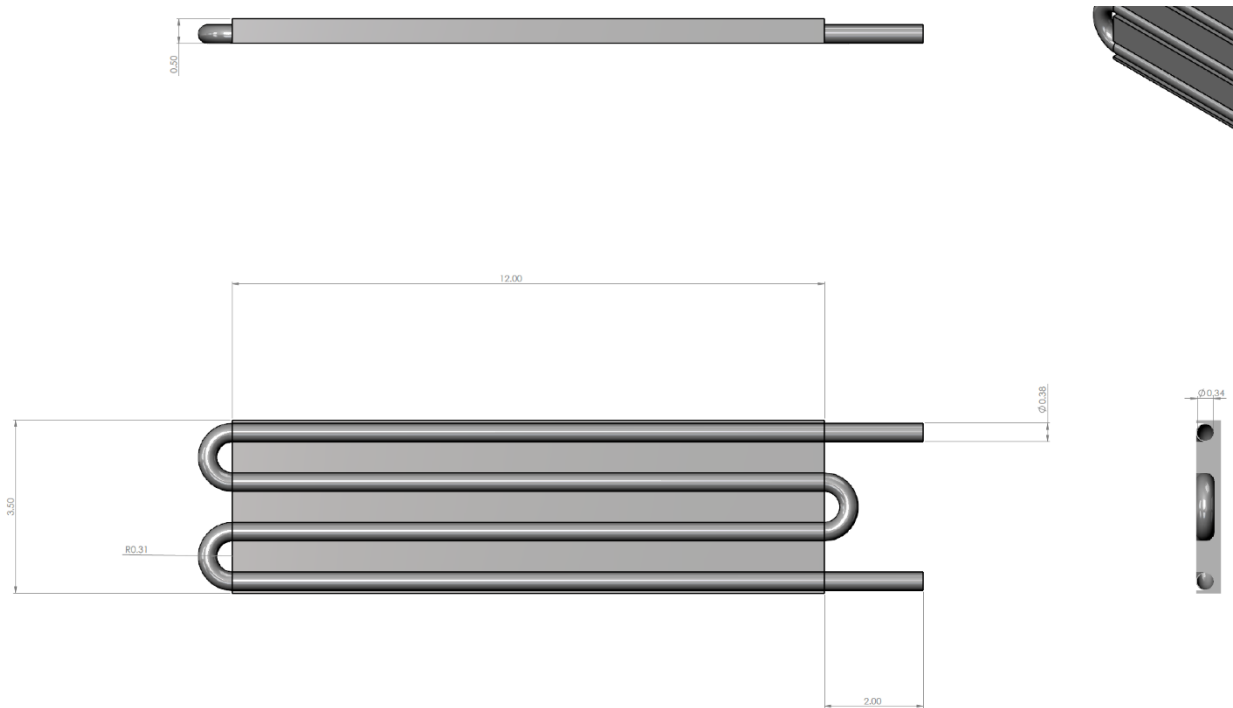


Figure 2. Zoomed in on dimensions of Cold Plate

Ansys
2024 R2

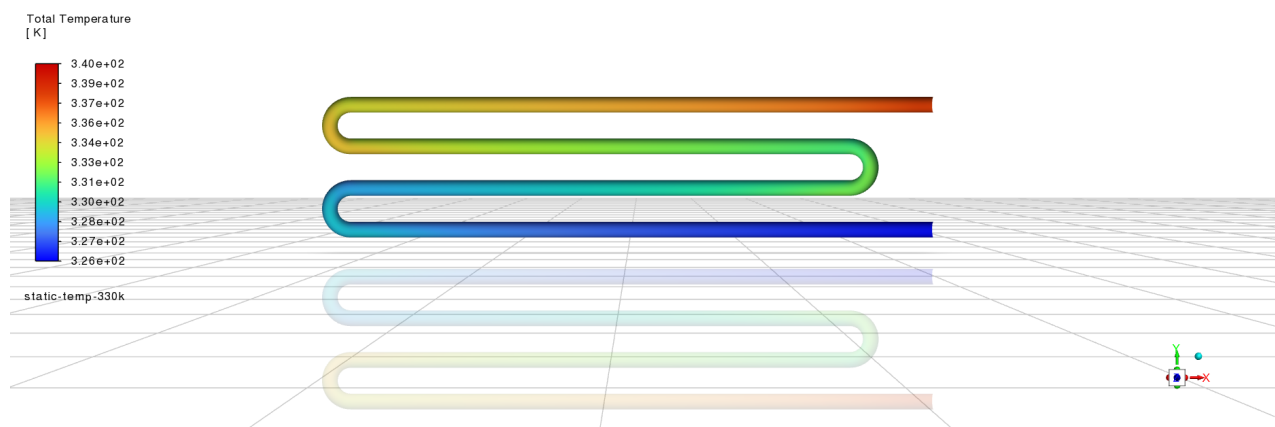


Figure 3. Static Temperature Contours – Outer Pipe

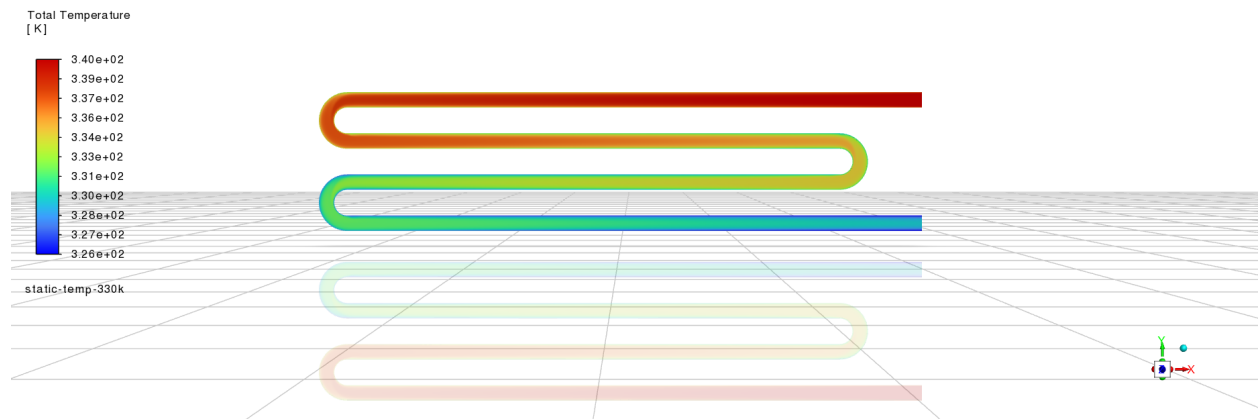


Figure 4. Static Temperature at Symmetry Plan (XY-axis)

Verification:

"Flux Report"	
Total Heat Transfer Rate	[W]
inlet	4950.3674
outerpipewall	-1327.9535
outlet	-3622.4139
wall-8	1327.9536
wall-8-shadow	-1327.9536
Net	-1.5473198e-05

Figure 5. Heat Flux

In the heat transfer of this cold plate heat exchanger, we see Figure 5 shows 4950.3674W at the inlet and -3622.4139W at the outlet – meaning the fluid is 1327.9535W less than at inlet. The rest of the heat transfer is through the pipe walling; thus, conservation of energy is satisfied.

"Flux Report"	
Mass Flow Rate	[kg/s]
inlet	0.028285083
outlet	-0.028285083
Net	-1.1238778e-10

Figure 6. Mass Flow Rate

As seen in Figure 6, the mass flow rate also checks out which is good because there's only an inlet and outlet, so there shouldn't be any lacking kg/s at the outlet.